

School of Computer Science and Artificial Intelligence

Lab Assignment # 16.2

Program : B. Tech (CSE)
Specialization : -
Course Title : AI Assisted Coding
Course Code : 23CS002PC304
Semester II
Academic Session : 2025-2026
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Enrollment No. : 2403A51L03
Batch No : 51
Date : 17/03/26

Task 1: (Design Database Schema for Student Management System)

Input:

Schema SQL

```
1 CREATE TABLE Students (  
2     student_id INT PRIMARY KEY,  
3     name VARCHAR(100),  
4     email VARCHAR(100)  
5 );  
6  
7 CREATE TABLE Courses (  
8     course_id INT PRIMARY KEY,  
9     course_name VARCHAR(100),  
10    credits INT  
11 );  
12  
13 CREATE TABLE Enrollments (  
14     enrollment_id INT PRIMARY KEY,  
15     student_id INT,  
16     course_id INT,  
17     FOREIGN KEY (student_id) REFERENCES Students(student_id),  
18     FOREIGN KEY (course_id) REFERENCES Courses(course_id)  
19 );  
20  
21 -- SHOW OUTPUT  
22 SHOW TABLES;  
23 DESCRIBE Students;  
24 DESCRIBE Courses;  
25 DESCRIBE Enrollments;
```

Output:

Results Copy as Markdown

Query #5 Execution time: 0.33ms

Field	Type	Null	Key	Default	Extra
student_id	int(11)	NO	PRI	null	
name	varchar(100)	YES		null	
email	varchar(100)	YES		null	

Query #6 Execution time: 0.32ms

Field	Type	Null	Key	Default	Extra
course_id	int(11)	NO	PRI	null	
course_name	varchar(100)	YES		null	
credits	int(11)	YES		null	

Query #7 Execution time: 0.37ms

Field	Type	Null	Key	Default	Extra
enrollment_id	int(11)	NO	PRI	null	
student_id	int(11)	YES	MUL	null	
course_id	int(11)	YES	MUL	null	

Explanation:

1. Defines structure of database using tables: Students, Courses, Enrollments.
2. Primary keys ensure unique identification of each record.
3. Foreign keys maintain referential integrity between related tables.
4. Follows normalization (1NF, 2NF, 3NF) to remove redundancy and ensure consistency.

Task 2: (Insert Sample Records)

Input:

 Database: MySQL v5.7
 Run Update Fork Load Example Star PRO

Query SQL PRO
Have any feedback?

```

1 INSERT INTO Students VALUES
2 (1, 'Arjun', 'arjun@gmail.com'),
3 (2, 'Meera', 'meera@gmail.com'),
4 (3, 'Rahul', 'rahul@gmail.com');
5
6 INSERT INTO Courses VALUES
7 (101, 'Database Systems', 4),
8 (102, 'Data Structures', 3),
9 (103, 'Operating Systems', 4),
10 (104, 'AI Fundamentals', 3);
11
12 INSERT INTO Enrollments VALUES
13 (1, 1, 101),
14 (2, 1, 102),
15 (3, 2, 103),
16 (4, 3, 101),
17 (5, 3, 104);
18
19 -- ☒ SHOW RESULT
20 SELECT * FROM Students;
21 SELECT * FROM Courses;
22 SELECT * FROM Enrollments;
    
```

Output:

Results

Copy as Markdown

student_id	name	email
1	Arjun	arjun@gmail.com
2	Meera	meera@gmail.com
3	Rahul	rahul@gmail.com

Query #5

Execution time: 0.07ms

course_id	course_name	credits
101	Database Systems	4
102	Data Structures	3
103	Operating Systems	4
104	AI Fundamentals	3

Query #6

Execution time: 0.06ms

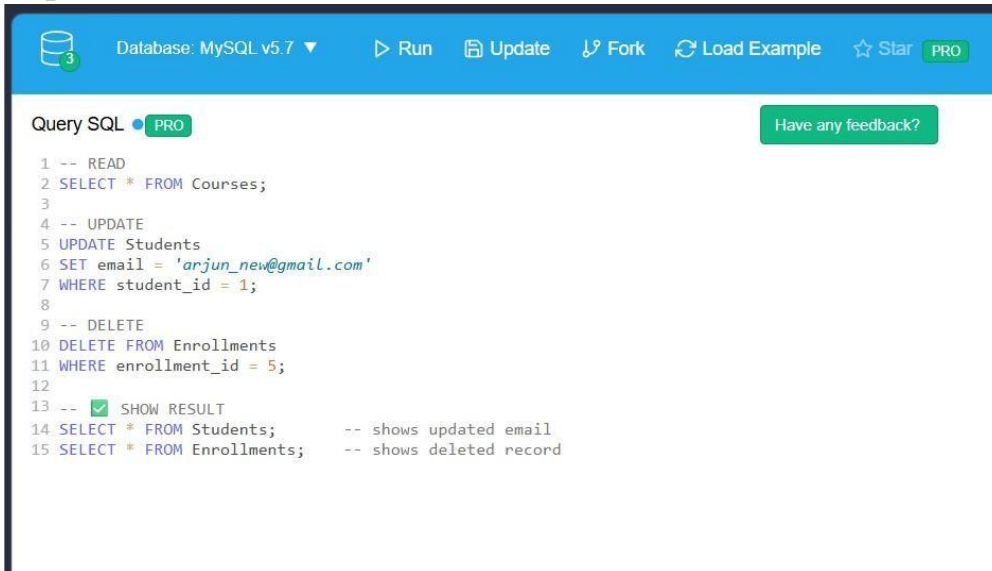
enrollment_id	student_id	course_id
1	1	101
2	1	102
3	2	103
4	3	101
5	3	104

Explanation:

1. Populates tables with initial data using INSERT statements.
2. Ensures relational links by matching student_id and course_id.
3. Helps in testing database functionality with sample dataset.
4. SELECT queries verify whether data insertion is successful.

Task 3: CRUD Operations

Input:



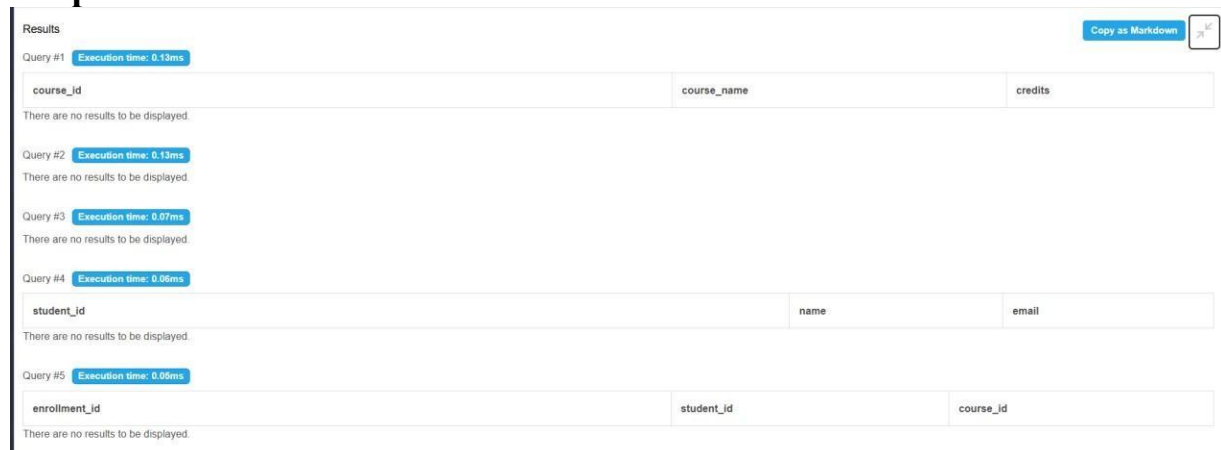
The screenshot shows a MySQL query editor with a blue header bar containing icons for database, run, update, fork, load example, star, and a PRO badge. The query text is as follows:

```

1 -- READ
2 SELECT * FROM Courses;
3
4 -- UPDATE
5 UPDATE Students
6 SET email = 'arjun_new@gmail.com'
7 WHERE student_id = 1;
8
9 -- DELETE
10 DELETE FROM Enrollments
11 WHERE enrollment_id = 5;
12
13 -- ☒ SHOW RESULT
14 SELECT * FROM Students;      -- shows updated email
15 SELECT * FROM Enrollments;  -- shows deleted record
  
```

A green button labeled "Have any feedback?" is located to the right of the query text.

Output:



The screenshot shows the results of five queries. Each query result is displayed in a table format with a header row and a message row indicating no results were returned.

Query #1	Execution time: 0.13ms
course_id	course_name
credits	
There are no results to be displayed.	

Query #2	Execution time: 0.13ms
There are no results to be displayed.	

Query #3	Execution time: 0.07ms
There are no results to be displayed.	

Query #4	Execution time: 0.06ms
student_id	name
email	
There are no results to be displayed.	

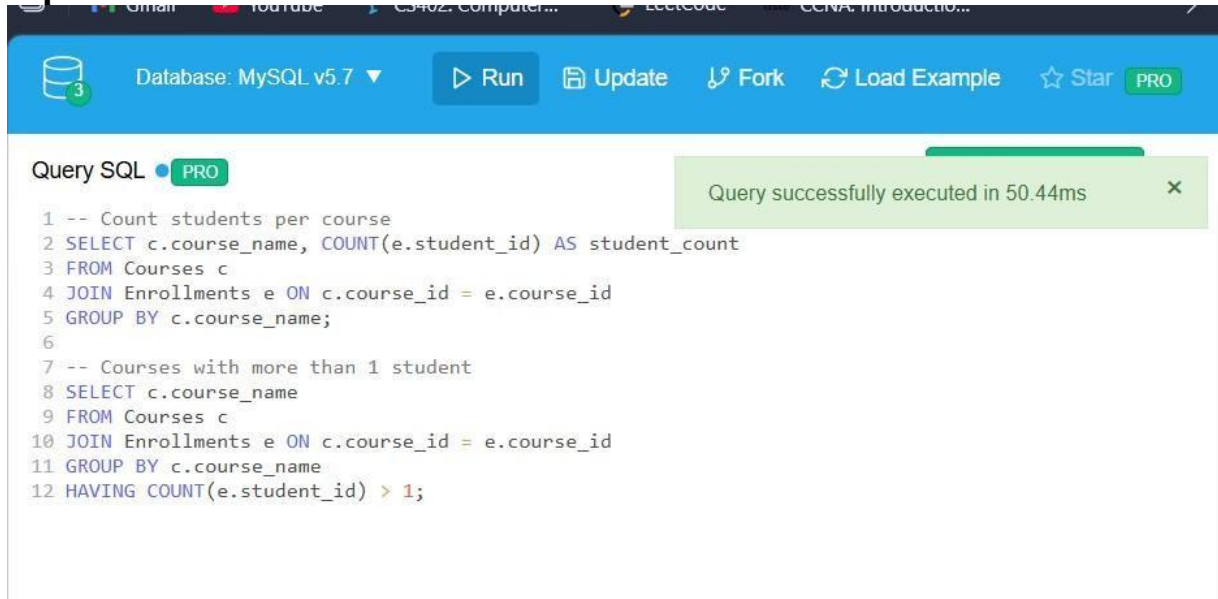
Query #5	Execution time: 0.06ms
enrollment_id	student_id
course_id	
There are no results to be displayed.	

Explanation:

1. SELECT retrieves data from tables for viewing records.
2. UPDATE modifies existing data like student email.
3. DELETE removes specific records such as enrollments.
4. Validates operations using SELECT to ensure correctness.

Task 4: Aggregate Queries

Input:



Database: MySQL v5.7

Run Update Fork Load Example Star PRO

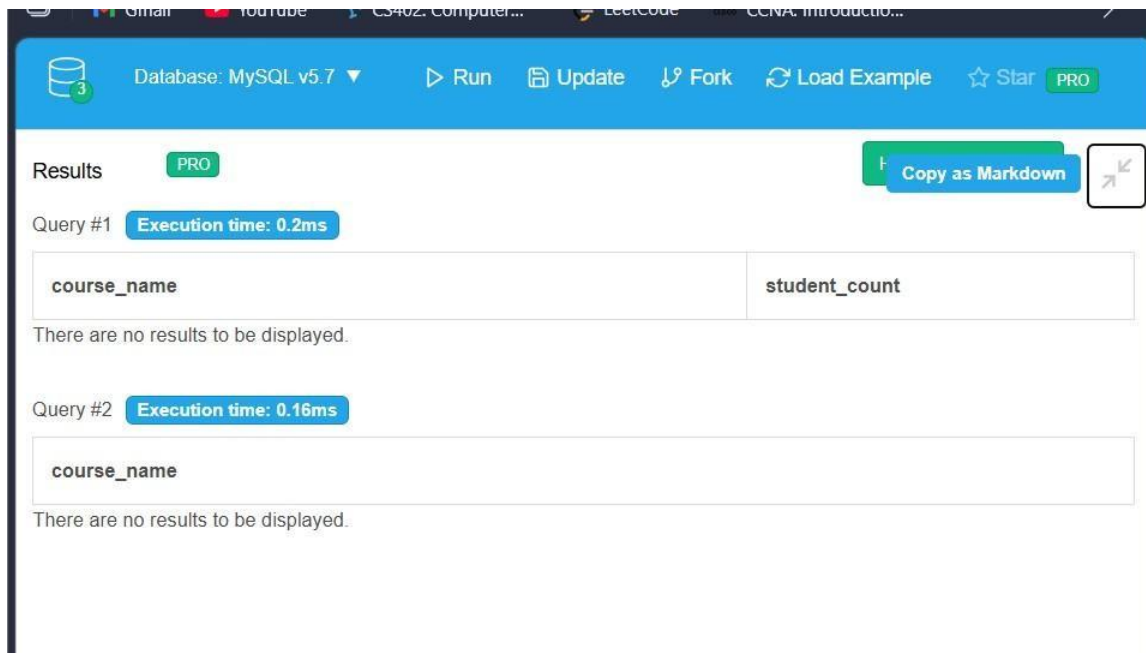
Query SQL PRO

Query successfully executed in 50.44ms

```

1 -- Count students per course
2 SELECT c.course_name, COUNT(e.student_id) AS student_count
3 FROM Courses c
4 JOIN Enrollments e ON c.course_id = e.course_id
5 GROUP BY c.course_name;
6
7 -- Courses with more than 1 student
8 SELECT c.course_name
9 FROM Courses c
10 JOIN Enrollments e ON c.course_id = e.course_id
11 GROUP BY c.course_name
12 HAVING COUNT(e.student_id) > 1;
  
```

Output:



Database: MySQL v5.7

Run Update Fork Load Example Star PRO

Results PRO

Copy as Markdown

Query #1 Execution time: 0.2ms

course_name	student_count
There are no results to be displayed.	

Query #2 Execution time: 0.16ms

course_name
There are no results to be displayed.

Explanation:

1. Uses functions like COUNT() to summarize data.
2. GROUP BY groups records based on course names.
3. HAVING filters grouped data (e.g., courses with >1 student).
4. Helps in analysis like enrollment trends per course.

Task 5: Query Optimization

Input:


Database: MySQL v5.7 ▼
Run
Update
Fork
Load Example
Star
PRO

Query SQL PRO

Query successfully executed in 8.19ms

```

1 -- Original
2 SELECT * FROM Students
3 WHERE student_id IN (
4     SELECT student_id FROM Enrollments WHERE course_id = 101
5 );
6
7 -- Optimized
8 SELECT s.*
9 FROM Students s
10 JOIN Enrollments e ON s.student_id = e.student_id
11 WHERE e.course_id = 101;
    
```

Output:


Database: MySQL v5.7 ▼
Run
Update
Fork
Load Example
Star
PRO

Results PRO
Copy as Markdown

Query #1 Execution time: 0.25ms

student_id	name	email
There are no results to be displayed.		

Query #2 Execution time: 0.14ms

student_id	name	email
There are no results to be displayed.		

Explanation:

1. Identifies inefficient queries (e.g., subqueries).
2. Rewrites queries using JOINS for better performance.
3. Ensures both original and optimized queries give same result.
4. Improves execution speed, especially for large datasets.